

EduPulse

Student Retention & Welfare Efficacy Tracker

From Messy Education Data to Actionable Insights

Data Dynamos

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The Problem: Fragmented Education Data

The State Education Department lacks a unified view of student welfare and retention-related risk across schools.



Student Attendance



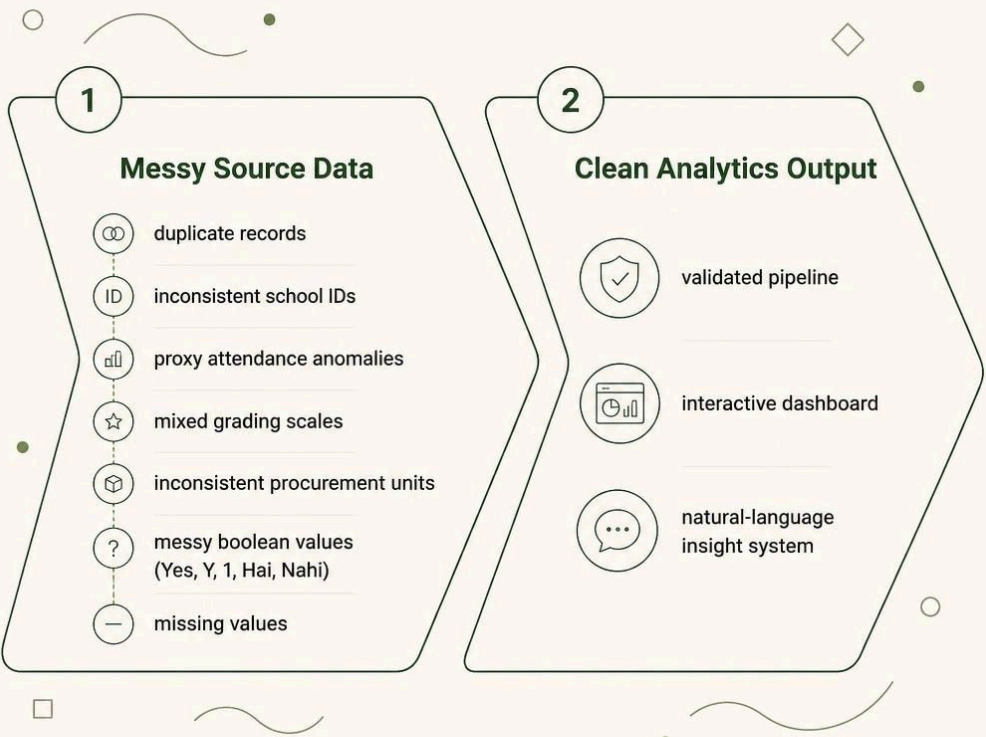
Academic Performance



Mid-Day Meal Procurement

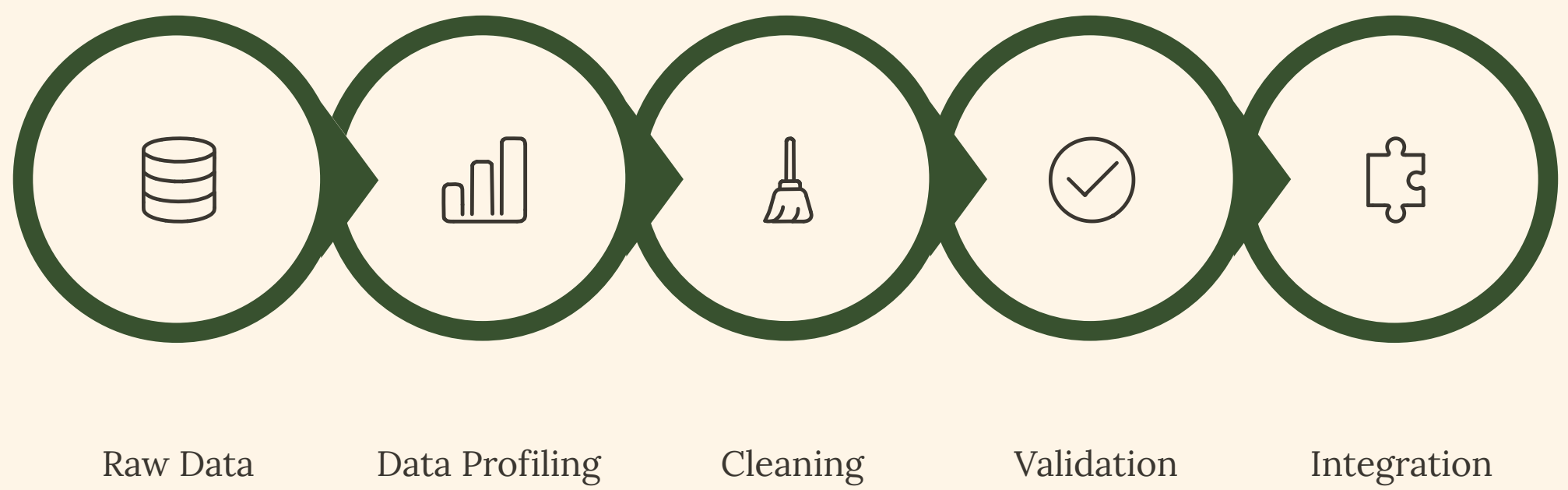


School Infrastructure



Source data contained duplicates, inconsistent IDs, proxy anomalies, mixed grading scales, and messy boolean values. EduPulse transforms these fragmented records into a validated analytics pipeline, interactive dashboard, and natural-language insight system.

Data Rescue: Turning Messy Data into Reliable Data



Each dataset was profiled, cleaned, and validated before integration into the school-level analytics layer.

Dataset	Raw Rows	Cleaned Rows	Rows Removed
School Master	618	600	18
Student Attendance	20,800	20,000	800
Test Scores	8,000	8,000	0
MDM Procurement	12,360	12,000	360
Infrastructure	3,150	3,000	150

📌 **600 schools** successfully integrated into the school-level analytics layer. Exact duplicates were removed. Missing values were not blindly imputed — legitimate missingness was retained and flagged.

Cleaning, Standardization & Validation



Attendance

- Standardized school IDs and dates
- $\text{Rate} = \text{present} \div \text{total} \times 100$
- 806 invalid records flagged
- 1,068 Sunday proxy records flagged

Records retained for audit.



Test Scores

- Percentage, Raw Marks, CGPA → common percentage scale
- 6,017 records with percentage values

Letter grades were not converted using an invented mapping.



MDM Procurement

- Grain names standardized
- KG, grams, 50 KG bags → KG
- Missing fields flagged, not imputed



Infrastructure

- Yes / Y / 1 / Hai / Haan → True
- No / N / 0 / Nahi / Broken → False

All school IDs validated against School Master.

Before → After: "sch-0140" → "SCH0140" · "40.7 KGS" → "40.7 KG" · "Hai" → TRUE · "43.2 / 50" → 86.4%

Analytics Layer: From Records to School-Level KPIs

600

Schools Analyzed

148K

Total Enrolled

148,083 students

81.66%

Avg Attendance

67.26%

Avg Test Score

3.12/5

Avg Facility Count

23.46

Avg Risk Indicator

Attendance Anomalies

- 1,068 Sunday 100% proxy-attendance records
- 806 invalid attendance records

Mid-Day Meal

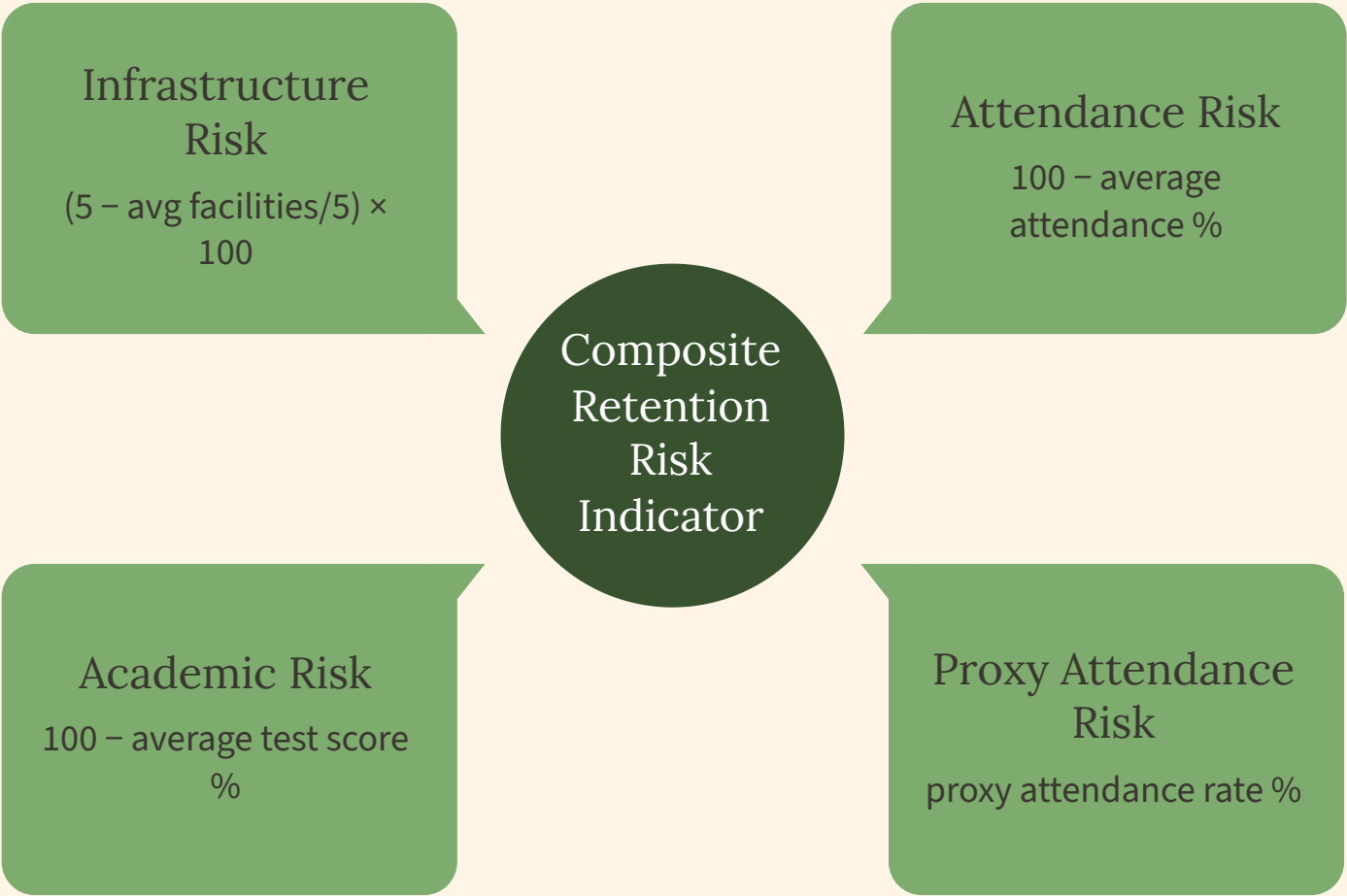
- 291,046.90 KG total grain quantity
- ₹27.92M total procurement cost
- 49.92% average paid percentage

Infrastructure Coverage

- 598 schools have infrastructure records
- 2 schools do not have infrastructure records

Retention Risk Indicator: Explainable Analytical Framework

❏ The dataset does not contain an actual dropout label. EduPulse does **not** predict dropout probability. The Retention Risk Indicator is an analytical measure to identify schools that may require attention.



Each component is calculated independently and combined as the mean of available risk components.

Highest Observed	Average	Lowest Observed
39.53	23.46	11.54

EduPulse Interactive Dashboard



Dashboard Modules

- Executive Command Center
- Academic & Attendance
- Infrastructure & Welfare
- School 360°
- Executive Memo
- Data Rescue & Audit
- Policy Impact Simulator
- Ask EduPulse AI

Attendance

Average attendance across schools = **81.66%**

Infrastructure

Electricity available in **405 of 541** schools with known status. Playground present in **260 of 526** schools.

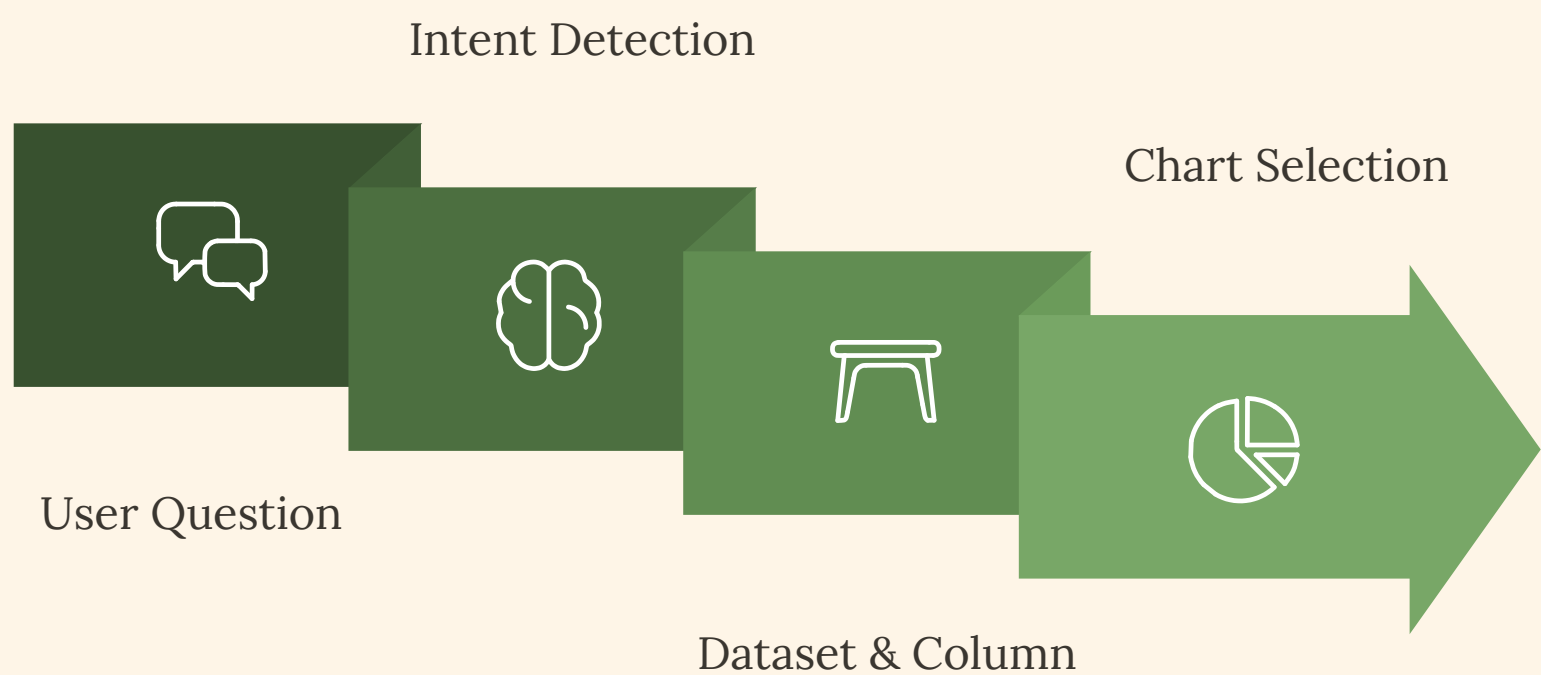
Academic Association

Attendance vs. test score Pearson correlation = **-0.002**, indicating essentially no linear association.

Electricity Comparison

With electricity: **67.45%** avg test score. Without: **67.29%**. Difference: **0.16** percentage points — an observed comparison, not a causal effect.

Ask EduPulse AI: Natural Language to Visual Insight



Users ask business questions in natural language; the system maps each question to an analytical operation and visualization.

"Show me the attendance trend over time."

→ Line Chart

"Compare average test scores by district."

→ Bar Chart

"Show the top 10 schools by retention risk."

→ Ranked Bar Chart

"Show the relationship between attendance and test scores."

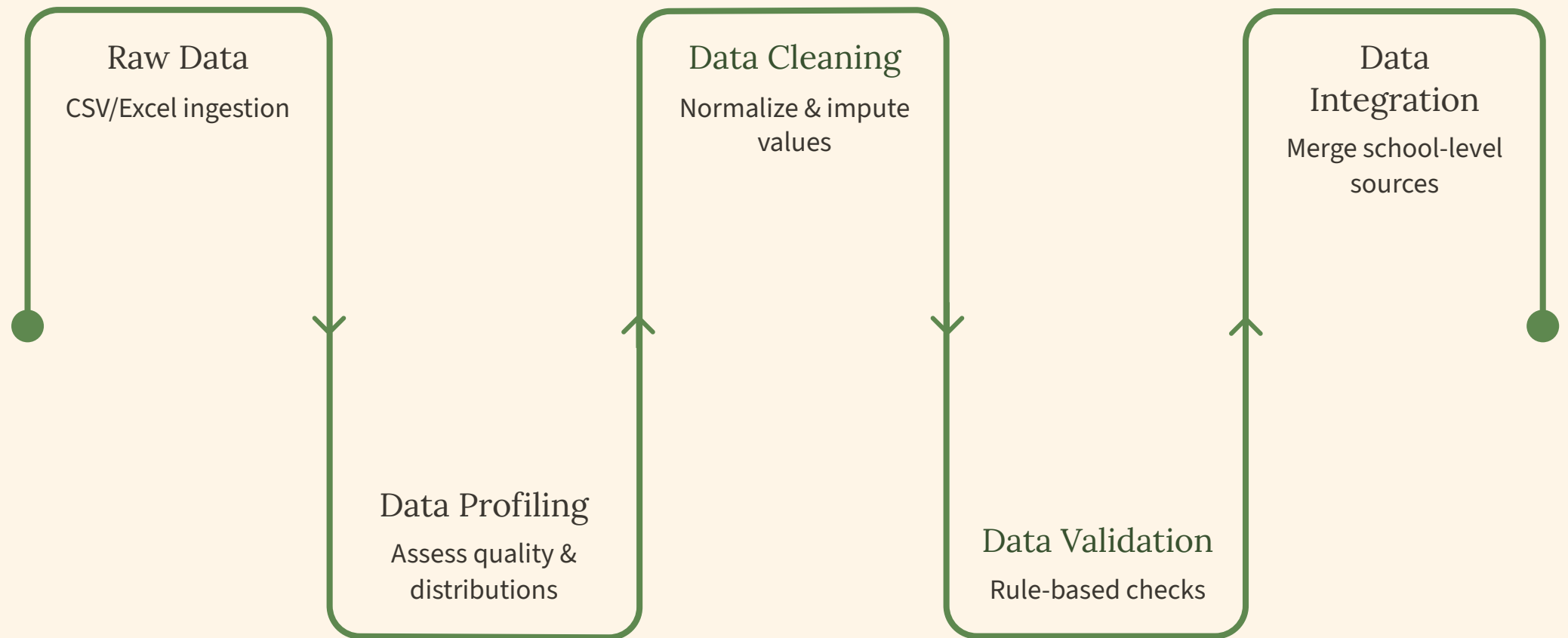
→ Scatter Plot

"Show schools with proxy attendance above 5%."

→ Anomaly Table / Chart

 **Graph-First Approach:** Trend → Line · Comparison → Bar · Relationship → Scatter · Ranking → Ranked · Anomaly → Table. Uses a **deterministic intent engine** — reproducible and auditable. Architecture supports future LLM integration.

EduPulse System Architecture



Technology Stack

- Python · Pandas · NumPy
- Streamlit · Plotly
- CSV / Excel data processing
- Natural-language intent detection
- Dynamic chart generation

Architecture Principles

- Reproducible
- Auditable
- Modular
- School-level integrated analytics
- Explainable business metrics

From Messy Data to Actionable Education Insights



Messy Data Rescue



Unified School Analytics



Interactive Dashboard



Explainable Risk Indicator



Graph-First AI Queries



Policy Simulator

What EduPulse Enables

- Identify schools requiring attention
- Monitor attendance anomalies
- Compare academic performance
- Identify infrastructure gaps
- Monitor Mid-Day Meal procurement
- Explore school-level retention risk indicators
- Ask questions using natural language

Future Scope

- Integrate historical dropout/retention labels
- Build validated ML-based risk models
- Add external LLM for broader queries
- Add automated alerts and intervention tracking
- Expand agent to additional education datasets

EduPulse transforms fragmented education data into a single, explainable view for monitoring student welfare and supporting data-driven decisions.